

Bispecific Fusion — Process & Assessment (Short Report)

Tandem single-chain bispecific: Aβ42 (anti-9CO4, Conf-1) arm + linker + TfR1 (anti-6WRV apical) BBB-shuttle arm. Source: ab42-binder-design repo, Stage-8 fusion outputs.

1. Construction — the linker step

The two validated arms were joined into single-chain tandem fusions across a combinatorial panel:

- **5 Aβ arms × 5 TfR1 arms × 2 domain orders × 5 linkers = 250 fusion candidates.**
- **Linkers:** GS3 (GGGGS×3, 15 aa), GS4 (20 aa), GS5 (25 aa) flexible; EAAAK rigid α-helical; PAPAP rigid Pro-rich.
- **Both domain orders:** Aβ–linker–TfR1 and TfR1–linker–Aβ.
- **Final size:** ~130–180 aa (median ~157 aa, ~17 kDa) — ~9× smaller than an IgG antibody.

2. Assessment performed — Stage 8 (computational)

Each of the 250 constructs was folded with ColabFold / AlphaFold2-ptm (1 model, 3 recycles, single-sequence MSA) and scored against three hard gates:

Gate	Threshold	Purpose
Aβ-arm pLDDT	≥ 80	Aβ arm folds correctly within the fusion
TfR1-arm pLDDT	≥ 80	TfR1 arm folds correctly within the fusion
inter-domain PAE	≥ 15	Domains remain independent — no inter-domain packing / paratope occlusion
pTM (soft)	≥ 0.60	Advisory only — low pTM expected for two independent domains

Result: 183 / 250 passed all hard gates (73%) → 20 designs selected as the synthesis panel (diversity-enforced: all 5 Aβ arms, all 5 TfR1 arms, balanced 10 Aβ-first / 10 TfR1-first).

3. Developability — what was actually assessed (verified in repo)

Developability WAS assessed — at two levels. It is computational/predictive; no experimental data yet.

(a) Arm-level — full biophysical panel (Stage 4). The individual Aβ-arm designs were run through the complete developability filter set and scored pass/fail:

Metric assessed	File / result
SAP aggregation score (sap_score, sap_per_residue)	stage4_results_recalibrated.csv
Buried unsatisfied H-bonds	stage4_results_recalibrated.csv
Net charge at pH 7.4	stage4_results_recalibrated.csv
Thermal-stability proxy (Tm_proxy)	stage4_results_recalibrated.csv
Polar contact-molecular-surface fraction	stage4_results_recalibrated.csv
Unpaired cysteine count	stage4_results_recalibrated.csv

Result: 23 of 62 arm designs passed all developability filters (pass_filters_1–6).

(b) Fusion-level — sequence-liability screen (Stage 9). The 20-design synthesis panel was screened for developability liabilities at the sequence level:

Property assessed	Detail
Isoelectric point (pI), MW	construct_pl, construct_mw_kda
Hydropathy (GRAVY), extinction coeff.	fusion_gravy, fusion_ext_coeff
Sequence liabilities	deamidation (NG), isomerization (DG/DP), N-glyc sequons, Met N-term oxidation
Severe flags / constraints	severe_flags, constraints_ok

Result: 0 of 20 fusions had severe flags; all 20 passed constraints (constraints_ok = True). Each carries only minor, common liabilities (a few deamidation/isomerization motifs) — none disqualifying.

What was NOT done. A full *structural* developability re-run (SAP / Tm / aggregation) on the assembled fusion constructs was not performed — the Stage-8 fusion files carry only fold + domain-independence gates, and Stage 9 adds sequence-liability screening, not structural SAP/Tm. No experimental developability (expression, SEC-MALS, aggregation, nanoDSF/CD melt) has been done — that is the next (wet-lab) step.

4. Accurate one-line summary

"We built 250 linker-fused bispecifics, fold-verified them (183 passed → 20-design panel), and screened the panel for developability: the individual arms passed a full biophysical panel (SAP, Tm-proxy, charge, buried-unsat, polar-CMS; 23/62), and the 20 fusions passed a sequence-liability screen (0 severe flags). Structural fusion-level SAP/Tm and experimental characterization are the next step."

Bottom line: 250 fusions constructed → 183 fold-verified → 20-design panel. Developability WAS assessed — a full biophysical panel on the arms (23/62 pass) and a sequence-liability screen on the 20 fusions (0 severe flags). Structural fusion-level SAP/Tm and wet-lab validation remain.